

Class 08: Breast Cancer

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Background

The goal of this mini-project is to explore a complete analysis using the unsupervised learning techniques covered in class. You'll extend what you've learned by combining PCA as a pre-processing step to clustering using data that consist of measurements of cell nuclei of human breast masses. This expands on our RNA-Seq analysis from last day.

FNA is a type of biopsy procedure where a very thin needle is inserted into an area of abnormal tissue or cells with a guide of CT scan or ultrasound monitors (Figure 1). The collected sample is then transferred to a pathologist to study it under a microscope and examine whether cells in the biopsy are normal or not.

For example radius (i.e. mean of distances from center to points on the perimeter), texture (i.e. standard deviation of gray-scale values), and smoothness (local variation in radius lengths). Summary information is also provided for each group of cells including diagnosis (i.e. benign (not cancerous) and malignant (cancerous)).

Data Import

Our data comes from the U. of Wisconsin Medical Center

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)

head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790
84348301	0.14250	0.28390	0.2414	0.10520
84358402	0.10030	0.13280	0.1980	0.10430
843786	0.12780	0.17000	0.1578	0.08089

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se	
842302	0.2419		0.07871	1.0950	0.9053	8.589
842517	0.1812		0.05667	0.5435	0.7339	3.398
84300903	0.2069		0.05999	0.7456	0.7869	4.585
84348301	0.2597		0.09744	0.4956	1.1560	3.445
84358402	0.1809		0.05883	0.7572	0.7813	5.438
843786	0.2087		0.07613	0.3345	0.8902	2.217

	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003		0.006193	25.38	17.33

842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67
843786	0.02165	0.005082	15.47	23.75
	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249
	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
842517	0.2416	0.1860	0.2750	
84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
84358402	0.4000	0.1625	0.2364	
843786	0.5355	0.1741	0.3985	
	fractal_dimension_worst			
842302	0.11890			
842517	0.08902			
84300903	0.08758			
84348301	0.17300			
84358402	0.07678			
843786	0.12440			

Q1. How many patients/samples are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```

B    M
357 212

```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
length(grep("_mean", colnames(wisc.df)))
```

```
[1] 10
```

There is a diagnosis column that is the clinician consensus that I want to exclude from any further analysis. We will come back later and compare our results to this diagnosis.

```
diagnosis = as.factor(wisc.df$diagnosis)
head(diagnosis)
```

```
[1] M M M M M M
Levels: B M
```

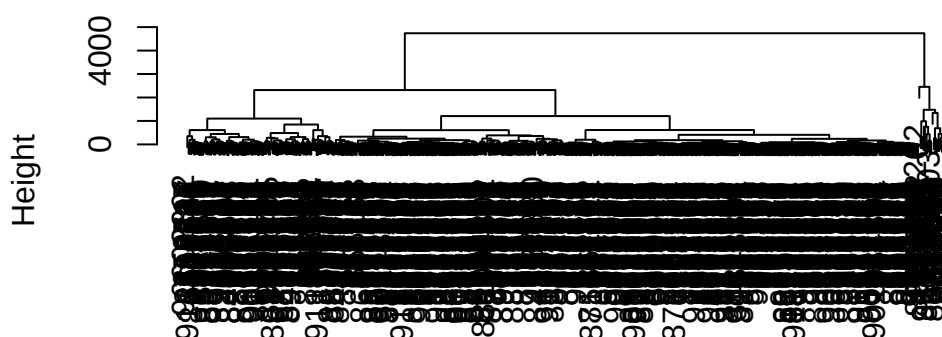
Now we can remove it from `wisc.df`.

```
wisc.data <- wisc.df[, -1]
```

Clustering

```
# kmeans(wisc.data, centers = 2)
hc <- hclust(dist(wisc.data), method = "complete")
plot(hc)
```

Cluster Dendrogram



```
dist(wisc.data)
hclust(*, "complete")
```

We can extract clusters from this rather poor dendrogram/tree with the `cutree()` function

```
grps <- cutree(hc, k=2)
```

How many individuals are in each cluster?

```
table(grps)
```

```
grps
  1  2
549 20
```

We can generate a cross-table that compares our cluster `grps` vector with `diagnosis` vector values.

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
B 357    0
M 192   20
```

Principle Component Analysis

The Importance of Data Scaling

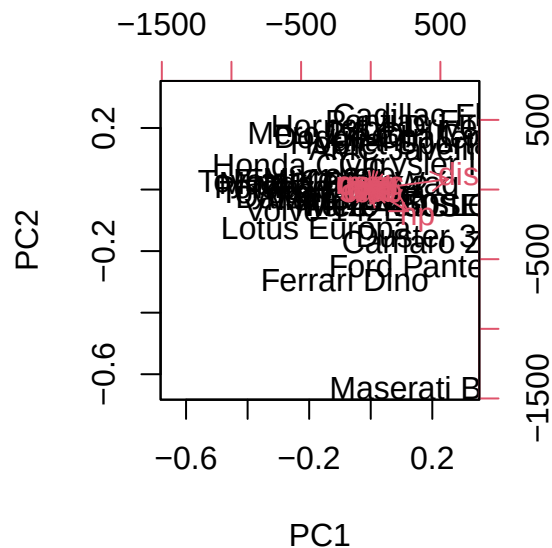
The main function for PCA in base R is `prcomp()`. It has a default input parameter of `scale = FALSE`.

```
head(mtcars)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

We could do a PCA of this data as is and it could be misleading.

```
pc <- prcomp(mtcars)
biplot(pc)
```



Lets look at the mean values of each columnn and their standard deviation.

```
# Check column means and standard deviations
colMeans(mtcars)
```

mpg	cyl	disp	hp	drat	wt	qsec
20.090625	6.187500	230.721875	146.687500	3.596563	3.217250	17.848750
vs	am	gear	carb			
0.437500	0.406250	3.687500	2.812500			

```
apply(mtcars,2,sd)
```

mpg	cyl	disp	hp	drat	wt
6.0269481	1.7859216	123.9386938	68.5628685	0.5346787	0.9784574
qsec	vs	am	gear	carb	
1.7869432	0.5040161	0.4989909	0.7378041	1.6152000	

We can “scale” this data before PCA to get a much better representation and analysis of all the columns.

```
mtscale <- scale(mtcars)
```

```
round(colMeans(mtscale))
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	0	0	0	0	0	0	0	0	0	0

```
apply(mtscale,2,sd)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	1	1	1	1	1	1	1	1	1	1

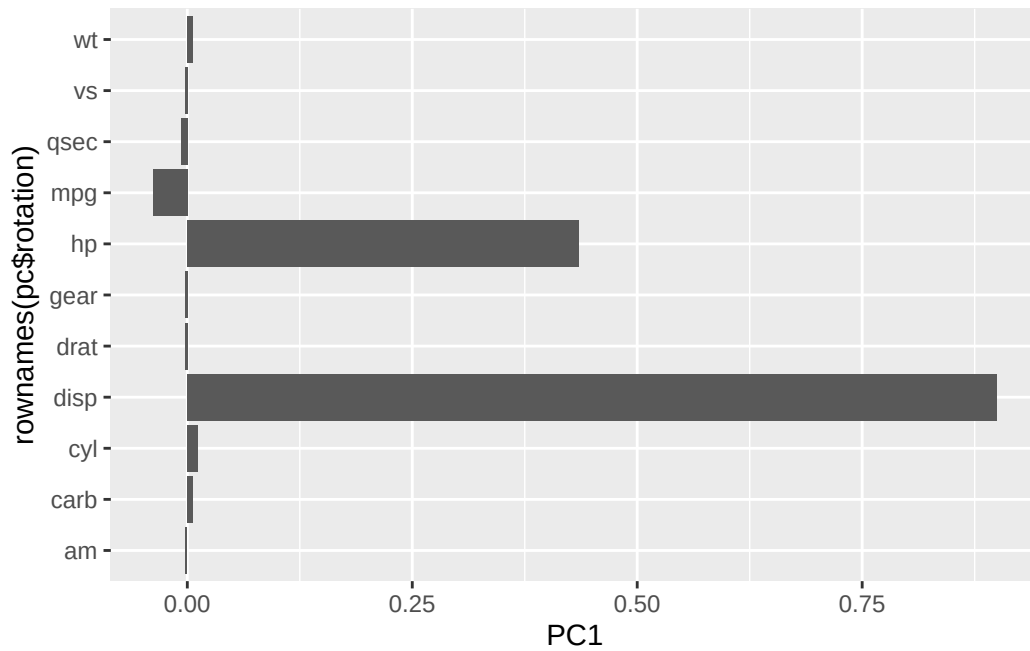
```
pc.scale <- prcomp(mtscale)
```

We can look at the two main results from PCA - the “PC plot” (aka the score plot, ordination plot, or PC1 vs PS2 plot), and the “loadings plot or how the original variables contribute to the new PCs.

A loadings plot of the unscaled PCA results

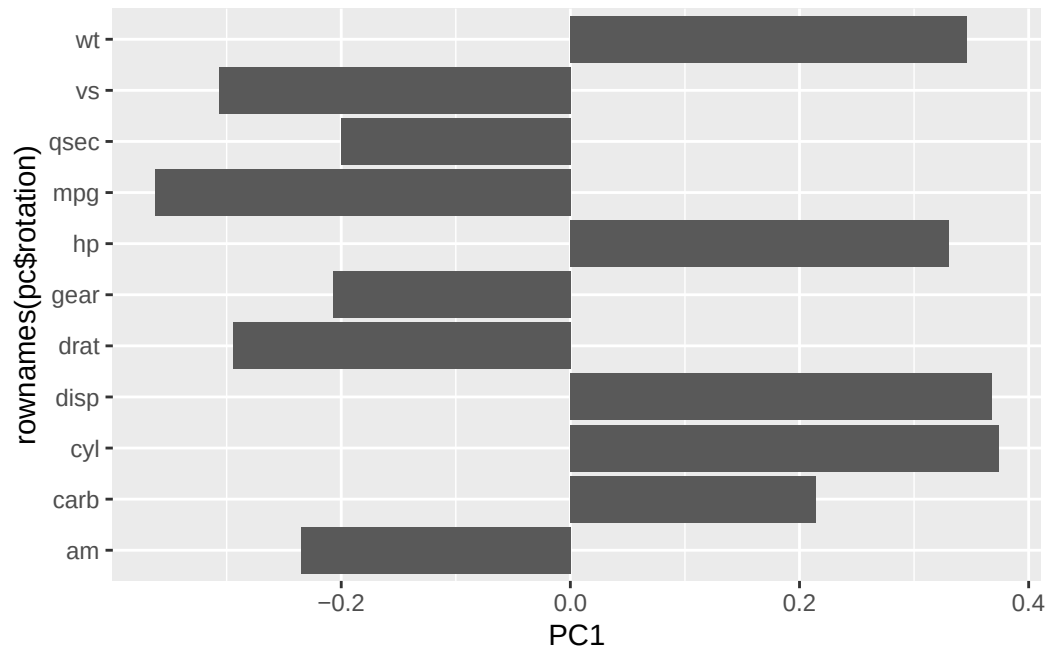
```
library(ggplot2)

ggplot(pc$rotation) +
  aes(PC1, rownames(pc$rotation)) +
  geom_col()
```



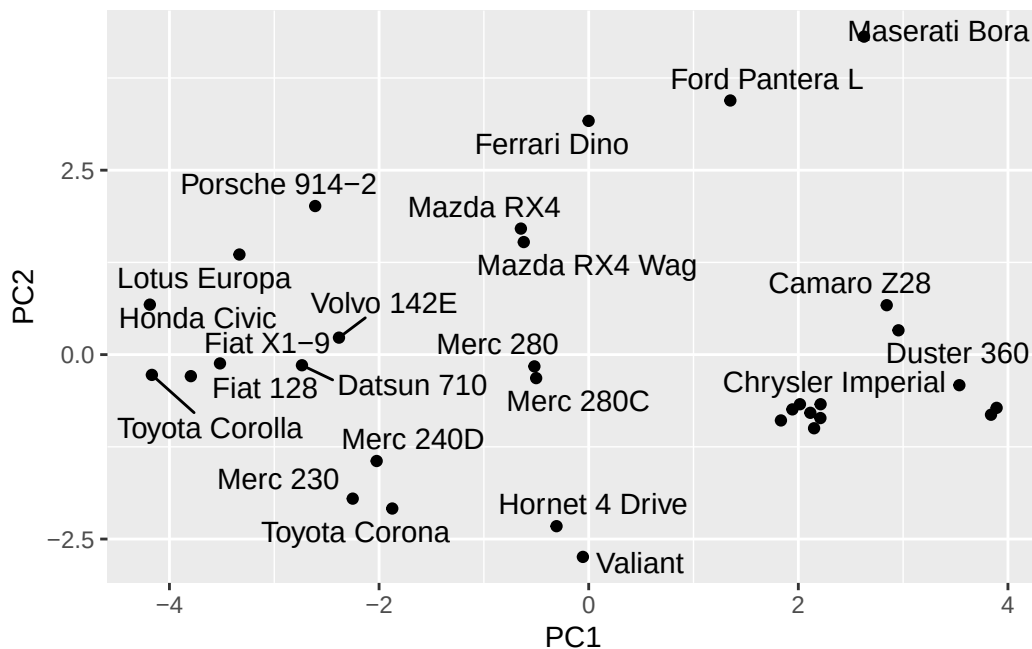
Loadings plot of the scaled data.

```
ggplot(pc.scale$rotation) +
  aes(PC1, rownames(pc$rotation)) +
  geom_col()
```

```
library(ggrepel)
ggplot(pc.scale$x) +
  aes(PC1, PC2, label=rownames(pc.scale$x)) +
  geom_point() +
  geom_text_repel()
```

Warning: ggrepel: 9 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Key point: In general, we will set `scale=TRUE` when we do PCA. This is not the default but probably should be...

We can check the SD and mean of the different columns in 'wisc.data1' to see if we need to scale - hint: we do!

PCA of wisc.data

```
wisc.pr <- prcomp(wisc.data, scale = TRUE)
```

To see how well PCA is doing here in terms of capturing the variance (or spread) in the data, we can use the `summary()` function.

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.7811	0.6310	0.5196	0.4177	0.3143	0.2108	0.1078
Proportion of Variance	0.0192	0.0160	0.0132	0.0105	0.0079	0.0053	0.0027
Cumulative Proportion	0.9293	0.9453	0.9585	0.9690	0.9769	0.9822	0.9849

Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

0.4427

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

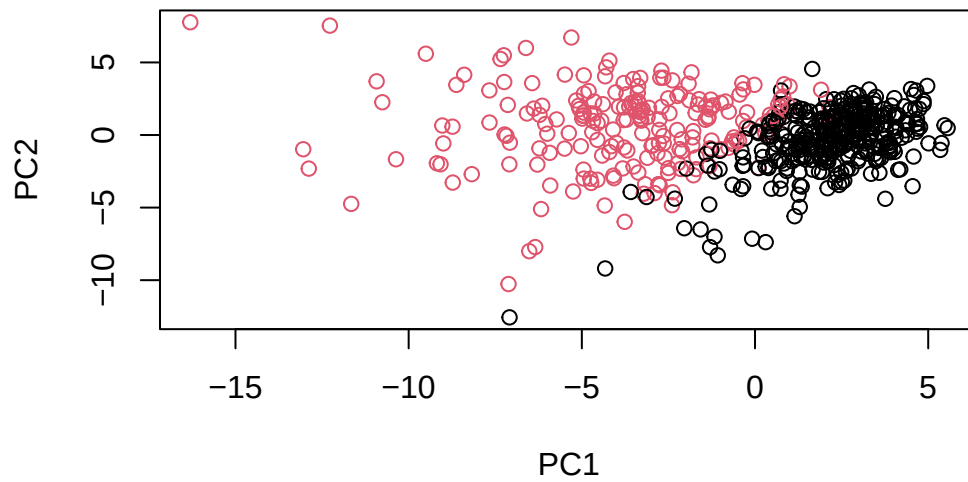
3

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7

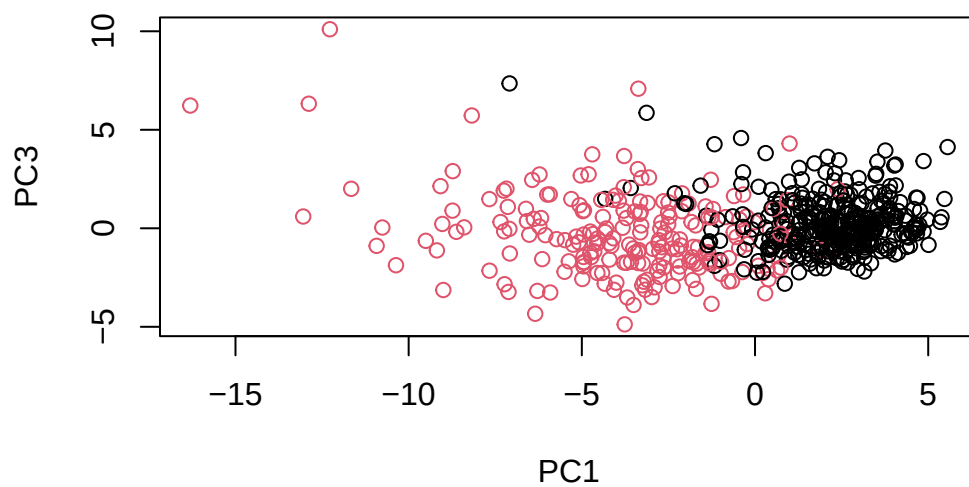
There are some challenges with running biplot.

```
biplot(wisc.pr)
```

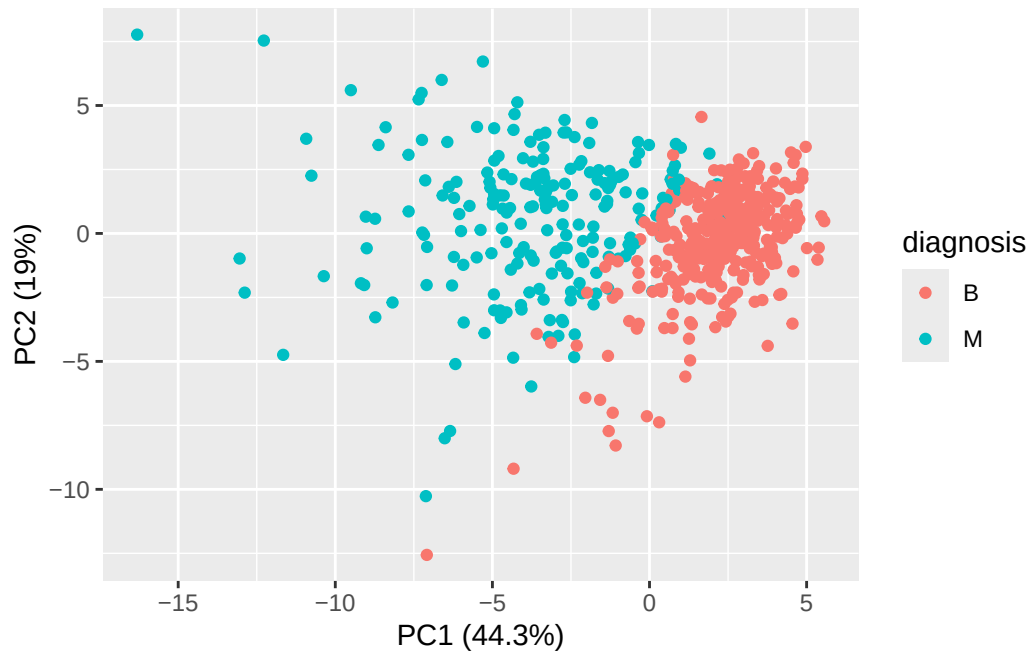
Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
plot(wisc.pr$x[,1], wisc.pr$x[,3], col = diagnosis,  
     xlab = "PC1", ylab = "PC3")
```



Lets make the main PC1 vs PC2 plot.

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC2, col = diagnosis) +  
  geom_point() +  
  xlab("PC1 (44.3%)") +  
  ylab("PC2 (19%)")
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`?

-0.26085376

Q10. What is the minimum number of principal components required to explain 80% of the variance in this data?

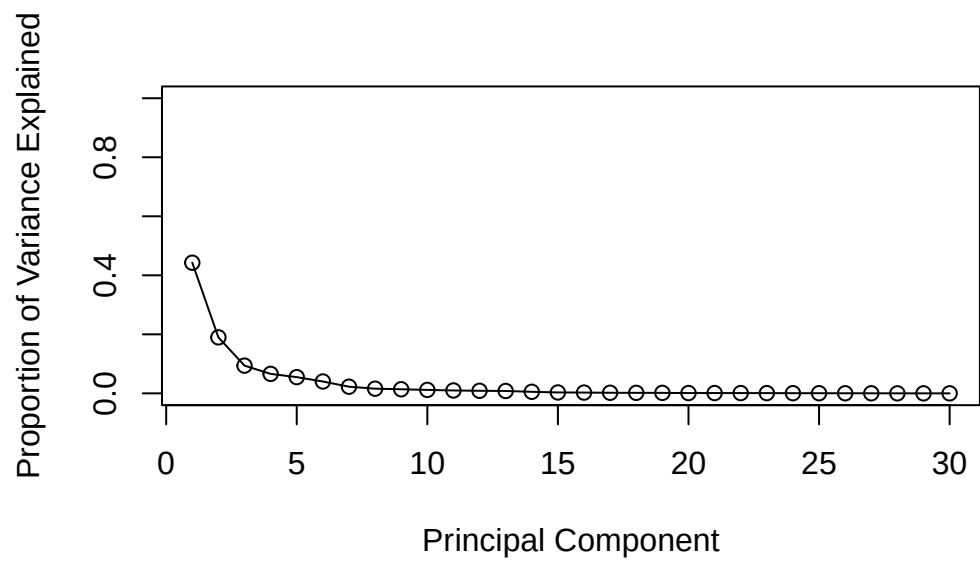
5

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

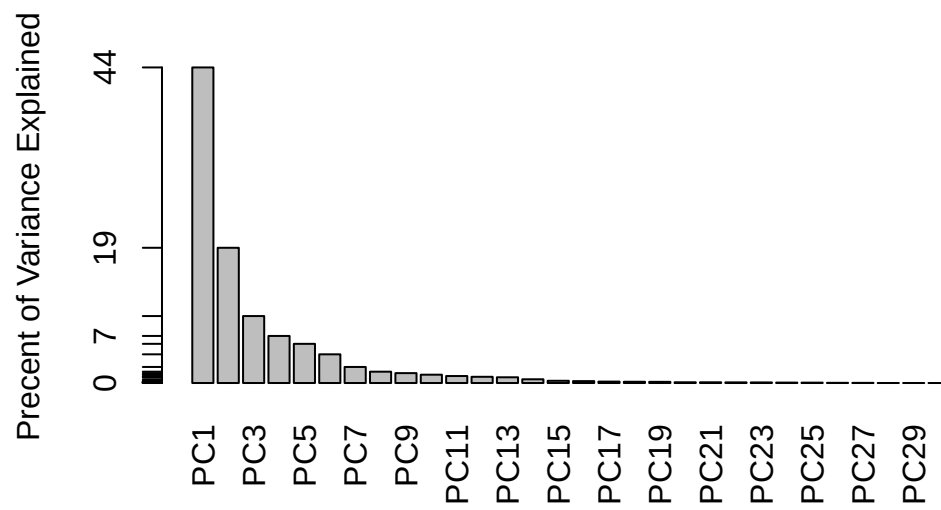
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(pve, xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Precent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```

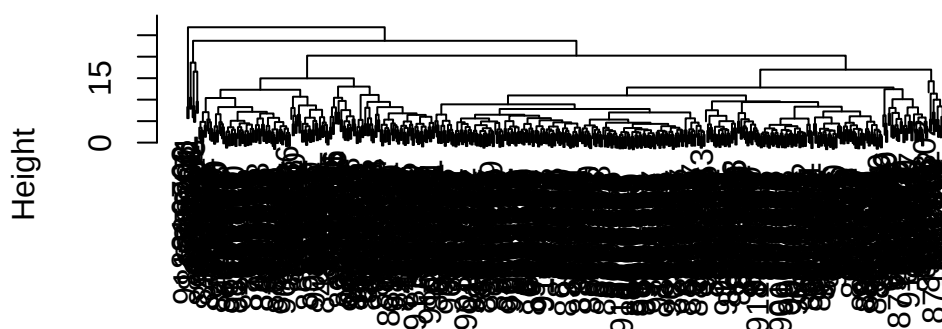
Hierarchical Clustering

Selecting number of clusters

```
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
```

```
wisc.hclust <- hclust(data.dist, method = "complete")
plot(wisc.hclust)
abline(wisc.hclust, col="red", lty=2)
```

Cluster Dendrogram



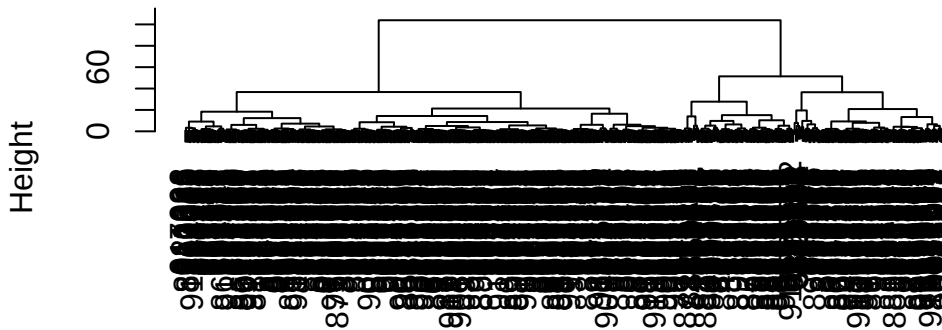
```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)  
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

K-means clustering

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:2]), method = "ward.D2")  
plot(wisc.pr.hclust)
```

Cluster Dendrogram



```
dist(wisc.pr$x[, 1:2])
hclust (*, "ward.D2")
```

We can “cut” this tree to yield our clusters (groups):

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
195 374
```

How do my cluster groups compare to the expert diagnosis?

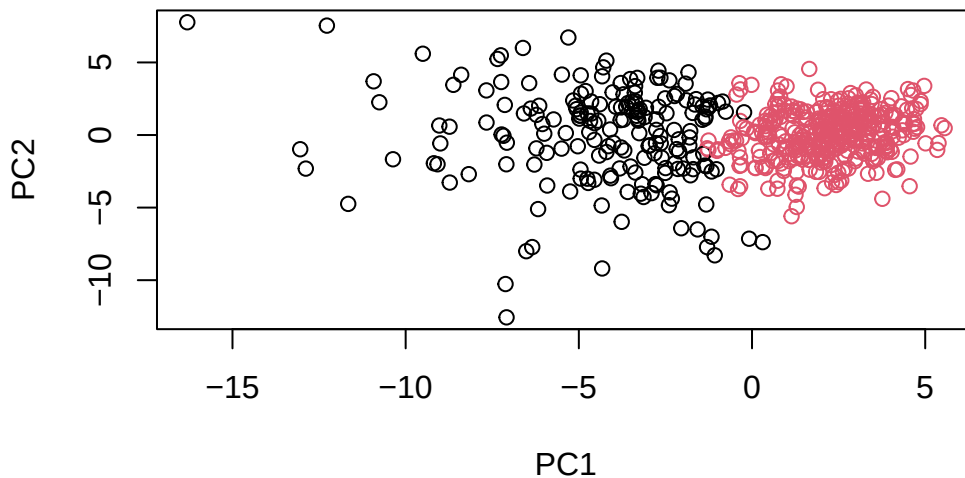
```
table(diagnosis, grps)
```

```

      grps
diagnosis  1   2
      B  18 339
      M 177  35

```

```
plot(wisc.pr$x[,1:2], col=grps)
```



Q15. How well does the newly created model with four clusters separate out the two diagnoses?

Compared to the expert diagnosis, the clusters formed by the model mostly overlap with each other. There are a couple instances where the model differs from the diagnosis but the vast majority are the same.

Q16. How well do the k-means and hierarchical clustering models you created in previous sections (i.e. before PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.km$cluster` and `wisc.hclust.clusters`) with the vector containing the actual diagnoses.

They did really badly. We do much better after PCA - the new PCA variables (what we call a basis set) give us a much better separation of M and B.

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031

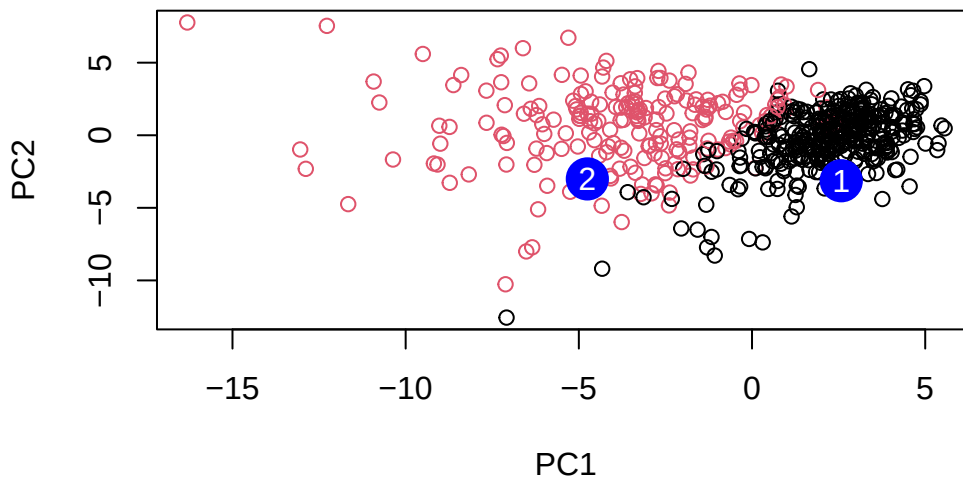
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029

	PC15	PC16	PC17	PC18	PC19	PC20
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500

	PC21	PC22	PC23	PC24	PC25	PC26
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238

	PC27	PC28	PC29	PC30
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820

```
plot(wisc.pr$x[,1:2], col=diagnosis)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q18. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize patient 2 since they are in the region identified as malignant.